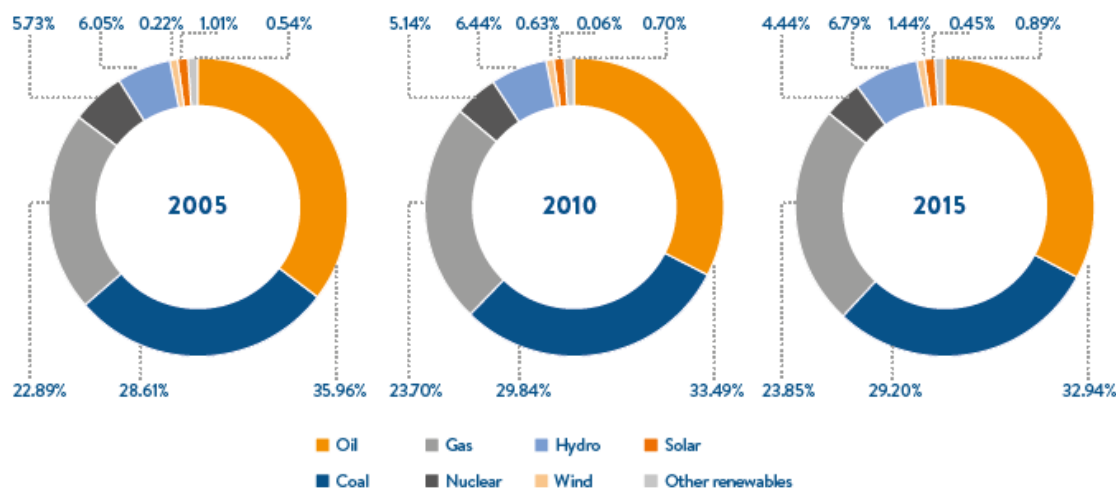


SUMMARY

The past 15 years have seen unprecedented change in the consumption of energy resources. Unexpected high growth in the renewables market, in terms of investment, new capacity and high growth rates in developing countries have changed the landscape for the energy sector. We have seen the growth of unconventional resources and improvements in technology evolution for all forms of energy resources. This has contributed to falling prices and the increased decoupling of economic growth and GHG emissions. Most countries have achieved a more diversified energy mix with a growth in community ownerships and an evolution of micro grids.

To better understand these unprecedented changes the 2016 World Energy Resources report highlights the key trends and identifies the implications for the energy sector.

FIGURE 1: COMPARATIVE PRIMARY ENERGY CONSUMPTION OVER THE PAST 15 YEARS



KEY FINDINGS

Solar

Global installed capacity for solar-powered electricity has seen an exponential growth, reaching around 227 GW_e at the end of 2015, producing 1% of all electricity used globally. The total capacity for solar heating and cooling in operation in 2015 was estimated at 406 GW_{th}.

As solar PV module prices have declined around 80% since 2007 (from ~ US\$4/W in 2007 to ~ US\$1.8/W in 2015), the cost associated with balancing the system represents the next great challenge for the Solar PV industry.

E-storage

E-storage has been characterised by rapid change, driven by reduced costs (especially batteries) and increased industry requirement to manage system volatility.

As of end-2015, the global installed storage capacity was 146 GW, consisting of 944 projects. There are already around 25 000 residential-scale units in Germany alone.

Bottom-up projections suggest a global storage market of 1.4 GW/y by 2020, with strong growth in electro-mechanical technologies in particular.

Marine

0.5 GW of commercial marine energy generation capacity is in operation and another 1.7 GW under construction, with 99% of this accounted for by tidal range.

The total theoretical wave energy potential is said to be 32 PWh/y, but is heterogeneous and geographically distributed, technology costs for marine energy are still very high, hindering deployment.

Uranium and Nuclear

Global uranium production increased by 40% between 2004 and 2013, mainly because of increased production by Kazakhstan, the world's leading producer.

As of December 2015, 65 nuclear reactors were under construction with a total capacity of 64 GW. Two-thirds (44) of the units under construction are located in three countries: China, India and Russia.

Currently there are more than 45 Small Modular Reactors designs under development and four reactors under construction.

Waste-to-Energy

Despite Waste-to-Energy (WtE) occupying less than 6% of the total waste management market, the global WtE market was valued at approximately US\$25 billion in 2015 and is expected to reach US\$36 billion by 2020, growing at CAGR of around 7.5 % between 2015 and 2020.

Hydropower

Hydropower is the leading renewable source for electricity generation globally, supplying 71% of all renewable electricity at the end of 2015. Undeveloped potential is approximately 10 000 TWh/y worldwide.

The global hydropower capacity increased by more than 30% between 2007 and 2015 accounting to a total of 1 209 GW in 2015, of which 145 GW is pumped storage.

Oil

Oil remained the world's leading fuel, accounting for 32.9% of global energy consumption. Crude oil prices recorded the largest percentage decline since 1986 (73%).

Roughly 63% of oil consumption is from the transport sector. Oil substitution is not yet imminent and is not expected to reach more than 5% for the next five years.

Unconventional oil recovery accounts for 30% of the global recoverable oil reserves and oil shale contains at least three times as much oil as conventional crude oil reserves, which are projected at around 1.2 trillion barrels.

Natural Gas

Natural gas is the second largest energy source in power generation, representing 22% of generated power globally and the only fossil fuel whose share of primary energy consumption is projected to grow.

Wind

Global wind power generation reached 432 GW in 2015, around 7% of total global power generation capacity (420 GW onshore, 12 GW offshore). A record of 63 GW was added in 2015 and total investment in the global wind sector was US\$109 billion in 2015.

Coal

Coal production decreased with 0.6% in 2014 and with a further 2.8% in 2015, the first decline in global coal production growth since the 1990s.

Coal still provides around 40% of the world's electricity. However, climate change mitigation demands, transition to cleaner energy forms and increased competition from other resources are presenting challenges for the sector.

Asia presents the biggest market for coal and currently accounts for 66% of global coal consumption.

CCS

CCS is an essential element of any low carbon energy future and industrial future, but policy is the main issue, not technology. The world's first large-scale application of CO₂ capture technology in the power sector commenced operation in October 2014 at the Boundary Dam power station in Saskatchewan, Canada.

There are 22 large-scale CCS projects currently in operation or under construction around the world, with the capacity to capture up to 40 million tonnes of CO₂ per year (Mtpa).

Geothermal

Geothermal global output is estimated to be 75 TWh for heat and 75 TWh for power, but is concentrated on geologic plate boundaries.

Bioenergy

Bioenergy is the largest renewable energy source with 14% out of 18% renewables in the energy mix and supplies 10% of global energy supply.